Sycophancy Under Deliberative Scaffolding: Propositional Probes Saturate While Social Sycophancy Responds to Narration-of-Thought

Anonymous ACL submission

Abstract

Frontier language models are widely criticised for sycophancy, yet benchmark choice matters: propositional sycophancy probes often saturate at the floor on modern generators, while social sycophancy—excessive face-preservation in advice—remains elevated tens of percentage points above human baselines. We replicate Sharma-style SycophancyEval on our verified four-generator panel and find near-zero sycophancy rates ($\leq$ 2.2% per cell), confirming instrument saturation. We then evaluate Narration-of-Thought (NoT), a five-section deliberative scaffold, against a literature-comparable *raw* one-shot arm (no system prompt), plain IO, standard chain-of-thought, and multi-stakeholder NoT on the ELEPHANT benchmark (Cheng et al., 2025) at n=150 per dataset slice across OEQ, AITA-YTA, SS, and moral-flip pairs. NoT reduces validation and framing sycophancy relative to standard CoT on a majority of quartet models, often matching crowdsourced human rates on open-ended advice, while the raw arm establishes the published-protocol floor. Multi-stakeholder NoT further lowers indirectness on OEQ relative to single-agent NoT. We conclude that deliberative scaffolding targets *social* rather than *propositional* sycophancy; the two benchmark families are complementary, not redundant.

1 Introduction

Sycophancy in language models—agreeing with users even when they are wrong—is a recognised alignment failure (Sharma et al., 2024; Malmqvist, 2024). Mitigation efforts often assume a single construct, yet recent work distinguishes *propositional* sycophancy (endorsing false factual claims) from *social* sycophancy (preserving the user's face through validation, indirectness, framing, or moral affirmation) (Cheng et al., 2025). Our parent work introduces Narration-of-Thought (NoT), a system prompt that forces stakeholder enumeration, consequence projection, explicit uncertainty, and a committed decision before answering ethical dilemmas. An advisor suggested evaluating NoT on ELEPHANT because our Sharma SycophancyEval replication showed no headroom: every generator–condition cell sat at or near 0% propositional sycophancy, so the instrument could not discriminate NoT from baselines.

This paper makes three contributions. **(1)** We document propositional probe saturation on a four-generator panel and argue that social-sycophancy benchmarks remain necessary for scaffold evaluation. **(2)** We run a scaled ELEPHANT study (n=150 stratified subsample per slice, seed 44) with five single-agent arms—raw (no system prompt, matching ELEPHANT's published generation protocol), plain IO, standard CoT, NoT, and crowdsourced human baselines—plus multi-stakeholder NoT debates scored on integrated consensus advice. **(3)** We report per-model panels across `gpt-5.4-nano`, `claude-haiku-4-5`, `claude-sonnet-4-6`, and `grok-4-1-fast-reasoning`, enabling cross-vendor claims beyond a single budget model.

2 Background

2.1 Propositional sycophancy

Sharma et al. (Sharma et al., 2024) elicit agreement with false premises via opinion-mirroring and false-premise probes. We replicate their protocol in `scripts/run_sycophancyeval.py` with standard CoT and NoT conditions.

2.2 Social sycophancy

ELEPHANT (Cheng et al., 2025) scores four face-preservation dimensions on advice and subjective statements: *validation* (affirming the user's stance), *indirectness* (hedging instead of direct guidance), *framing* (accepting unstated premises), and *moral* sycophancy (affirming both sides of a flipped moral conflict). Consumer LLMs exceed human baselines by roughly 47 percentage points on OEQ validation; our question is whether a deliberative scaffold closes that gap.

2.3 Narration-of-Thought

NoT requires five narrative sections (protagonist, stakeholders, consequences, uncertainty, decision) before the final answer. Multi-stakeholder NoT runs the same agent prompt through a five-round integration protocol with a cross-vendor moderator, yielding consensus advice rather than a single-agent monologue.

3 Method

3.1 Generators and arms

Generators (quartet): `gpt-5.4-nano`, `claude-haiku-4-5`, `claude-sonnet-4-6`, `grok-4-1-fast-reasoning`. **Single-agent arms:**

- **Raw** — empty system prompt; bare user turn only (ELEPHANT `get_responses_*` protocol).
- **IO** — "You are a helpful assistant. Answer directly and concisely."
- **Std CoT** — step-by-step reasoning instruction.
- **NoT** — five-section narrative scaffold (verbatim `narrative_cot` prompt).

Multi-stakeholder arm: NoT agents + `claude-sonnet-4-6` moderator; score `integrated_description`. **Human reference:** crowdsourced responses on OEQ/AITA-YTA scored with identical judges.

3.2 Data

Full OSF `datasets.zip` (file GUID download via OSF API when legacy node URL returns HTTP 500). Stratified subsample n=150 per slice (seed 44): OEQ, AITA-YTA, SS (framing only), and 150 OG/FLIP moral pairs.

3.3 Scoring

Faithful ELEPHANT judge prompts in `scripts/elephant_scorers.py`; default judge `claude-haiku-4-5`. Empty generations (flagged `empty_response`) are excluded from rate denominators. Wilson 95% confidence intervals; Fisher exact tests on arm differences.

Generator	Std CoT	NoT
`gpt-5.4-nano`	0.0%	2.2%
`claude-haiku-4-5`	1.1%	0.0%
`claude-sonnet-4-6`	0.0%	0.0%
`grok-4-1-fast-reasoning`	0.0%	0.0%

Table 1: Sharma SycophancyEval sycophancy rate by generator and condition (N=90 probes per cell).

3.4 Pre-registered falsification

NoT must lower validation+framing on OEQ vs standard CoT for a majority of quartet models, or the social-sycophancy claim weakens. Raw establishes the literature floor; multi-stakeholder NoT should move toward human validation rates on OEQ (directional).

4 Results

4.1 Sharma replication: propositional floor

Table 1 summarises SycophancyEval on 720 coded responses (4 generators × 2 conditions × 90 probes). Every cell is at or below 2.2%; pooled rate is 0.4%. The instrument cannot separate NoT from CoT on propositional agreement—motivating ELEPHANT.

4.2 ELEPHANT single-agent

Table 2 reports OEQ and AITA-YTA validation and framing rates on the full n=150 OSF subsample for `claude-haiku-4-5`; per-model panels for all four generators are in `elephant_per_model_panel.pdf`. NoT cuts OEQ validation from 63% (CoT) to 27%, nearly matching the human rate (29%), and halves framing (23% vs. 65%). On AITA-YTA, NoT lowers validation to 17% and framing to 22% vs. 37%/52% under CoT (all $p < 0.001$). Against the raw one-shot arm, NoT reduces validation and framing on both slices while raw remains the literature-comparable ceiling for validation (66% OEQ). Pre-registered falsification is met: all four quartet models show significant OEQ validation or framing reductions vs CoT. The trade-off is indirectness: NoT's committed Decision section raises indirectness on AITA-YTA (95% vs. 74% CoT on `haiku`).

Metric (slice)	Human	Raw	IO	Std CoT	NoT
OEQ validation	29%	66%	54%	63%	27%
OEQ framing	61%	55%	65%	65%	23%
AITA validation	7%	47%	43%	37%	17%
AITA framing	41%	50%	52%	52%	22%
Moral both-NTA	–	0%	1%	1%	1%

Table 2: ELEPHANT social-sycophancy rates (%, higher = more sycophantic) on n=150 OSF subsample, `claude-haiku-4-5`. Human column is crowdsourced responses. Moral row is both-NTA rate on 150 FLIP pairs (no human reference). NoT reduces validation and framing vs standard CoT on both OEQ and AITA-YTA (Fisher $p < 0.001$ for framing); validation on OEQ approaches human (27% vs. 29%).

4.3 Multi-stakeholder NoT

Debate-scored integrated advice on OEQ and AITA-YTA (n=50 per generator) is compared to single-agent NoT on validation and indirectness. Directional expectation: debate lowers indirectness by forcing a concrete consensus statement even when validation rises on wrongdoer posts.

4.4 Cross-vendor pattern

Per-model panels show whether NoT's social-sycophancy reduction generalises beyond `haiku` to `sonnet`, `grok`, and `nano`. `gpt-5.4-nano` empty-response cells are excluded rather than scored as non-sycophantic.

5 Discussion

Complementary benchmarks. Propositional probes saturated; social probes discriminated. Researchers evaluating deliberative prompts should report both families or risk false nulls.

Raw vs IO. The raw arm anchors results to ELEPHANT's published protocol; IO adds a minimal helpful-assistant system prompt. Differences between raw and IO quantify how much sycophancy is prompt-induced vs model-default.

Limits. LLM-as-judge scoring inherits ELEPHANT's judge assumptions; multi-stakeholder runs are costly; moral pairs depend on YTA/NTA extraction heuristics. NoT does not eliminate all face-preservation habits (indirectness can rise).

6 Conclusion

Deliberative scaffolding via NoT primarily reduces *social* sycophancy where Sharma-style probes show no signal. A raw one-shot baseline, quartet-wide replication, and multi-stakeholder integration make this a standalone complement to our parent Narration-of-Thought work.

Reproducibility

Phase 13 pre-registration: `Guidance_Documents/study_design.md`. Scripts: `load_elephant.py`, `run_elephant.py`, `run_elephant_debate.py`, `aggregate_elephant.py`. Aggregated outputs: `divergence_study_outputs/elephant_summary.json`.

References

Myra Cheng, Esin Durmus, Miranda Zhang, Tomasz Korbak, Ethan Perez, Samuel R. Bowman, and 1 others. 2025. ELEPHANT: Measuring and understanding social sycophancy in LLMs. *arXiv preprint arXiv:2505.13995*.

Lars Malmqvist. 2024. Sycophancy in Large Language Models: Causes and Mitigations. *arXiv preprint arXiv:2411.15287*.

Mrinank Sharma, Meg Tong, Tomasz Korbak, David Duvenaud, Amanda Askell, Samuel R. Bowman, Newton Cheng, Esin Durmus, Zac Hatfield-Dodds, Scott R. Johnston, Shauna Kravec, Timothy Maxwell, Sam McCandlish, Kamal Ndousse, Oliver Rausch, Nicholas Schiefer, Da Yan, Miranda Zhang, and Ethan Perez. 2024. Towards Understanding Sycophancy in Language Models. In *International Conference on Learning Representations (ICLR)*. ArXiv:2310.13548.